CS 385: Software Engineering Term Project

Lost & Found

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1. Chapter 1

System descriptions

The primary goal of our platform is to help students locate their lost property through a specific website. The platform offers a number of useful features, including reporting lost or found property, assisting in the recovery of lost property, and informing users of the progress of their requests.
The system also makes it easy for students to manage and modify their accounts. Additionally, it can continuously monitor each reported item until it is found and subsequently deleted.

The development Process model used

The model we will use for our system is the waterfall model, as we have clearly understood the requirements and it fits our needs.
The requirements are defined, clarifying the objective and reducing the possibility of confusion over details.
The waterfall structure is simple and maintains organization through proper processes.
In addition, it is not possible to move to the next stage until the previous one is completed.

System Requirements

Functional Requirements

User Requirements

1. The user shall be able to register a new account
2. The user shall be able to login to the system using their credentials 3. The user shall be able to change their password if needed
4. The user shall be able to browse the platform.
5. The user shall be able to report a lost item.
6. The user shall be able to upload images.
7. The user shall be notified when someone claims their reported item.
8. The user shall be able to mark an item as claimed.
9. The user shall delete items when returned.

System Requirements

1. The user shall be able to register a new account.
   1.1 The system shall provide a registration form for new users to enter their personal information like name, email and password.
   1.2 The system shall validate the entered information and if it's valid it will create a new account.
   1.3 The system shall notify the user if the registration was successful.

2. The user shall be able to login to the system using their credentials.
   2.1 The system shall provide a login interface where users can enter their username or email and password.
   2.2 The system shall authenticate the credentials to make sure its valid.
   2.3 The system shall provide an error message if the credentials were invalid.

3. The user shall be able to change their password if needed.
   3.1 The system shall provide an option to reset the password using a verified email which the users provided in the registration form.

3.2 The system shall allow the users to change their password after authentication.
3.3 The system shall notify the users if the password change was successful.

4. The user shall be able to browse the platform.
   4.1 The system shall allow users to view all reported lost and found items without the need to log in.
   4.2 The system shall display items in an organized layout with item name, image, description, and location.
   4.3 The system shall provide search functionality that allows users to find specific items using keywords.
   4.4 The system shall include filters (such as category, date, or location) to refine search results.
   4.5 The system shall display both lost and found items in separate sections for easy navigation.
   4.6 The system shall allow users to click on an item to view more detailed information.
   4.7 The system shall automatically update the displayed list when filters or search terms are changed.
   4.8 The system shall ensure that only active (not claimed or deleted) items are visible to users.
   4.9 The system shall support pagination or scrolling to view additional items smoothly.
   4.10 The system shall display a message when no matching items are found.

5. The user shall be able to report a lost item
   5.1 The system shall provide a form for users to fill out when reporting a lost item, including the description, location, time, and contact details.
   5.2 The system shall allow users to upload images of the lost item when filling out the report.
   5.3 The system shall store the lost item details in the database after submission.
   5.4 The system shall notify the user when the lost item report is successfully submitted.
   5.5 The system shall allow users to edit or cancel the reported item within a specific time frame after submission.

6. The user shall be able to upload images.
   6.1 The system shall allow users to upload the image of the lost and found item when filling out the report.
   6.2 The system shall store the image of the item in the database and should be easy for users to find the image when filtering the items in the system.

7. The user shall be notified when someone claims their reported item.
   7.1 The system shall automatically create a notification once another user claims the item.
   7.2 The notification shall be sent to the original reporter of the lost item.
   7.3 Notifications shall appear in the user's account under a "Notifications" section.
   7.4 The system may also send the user an email or SMS alert if they have this option enabled.
   7.5 The notification message shall include details such as the item name, claim date, and claimant information.
   7.6 The user shall be able to view and mark notifications as read.
   7.7 The system shall keep a record of all notifications in the database.
   7.8 The system shall prevent duplicate or repeated notifications for the same event.

8. The user shall be able to mark an item as claimed.

8.1 The system shall provide a button or option for users to mark an item as claimed from the item's report page.
8.2 The system shall ensure that only the user who reported the item (or authorized personnel) can mark the item as claimed.
8.3 The system shall update the item's status to "claimed" in the database once marked.
8.4 The system shall notify the user that the item has been successfully marked as claimed.
8.5 The system shall allow users to unmark the item as claimed, with confirmation before doing so.
8.6 The system shall notify all users when the item is marked as claimed and remove it from the lost items list.
8.7 The system shall store the date and time when the item was marked as claimed.

9. The user shall delete items when returned.
   9.1 The system shall allow the reporting users to delete the reported item in their account.
   9.2 The system should confirm the deletion before removing the item in case of deleting it by mistake.

Non-Functional Requirements

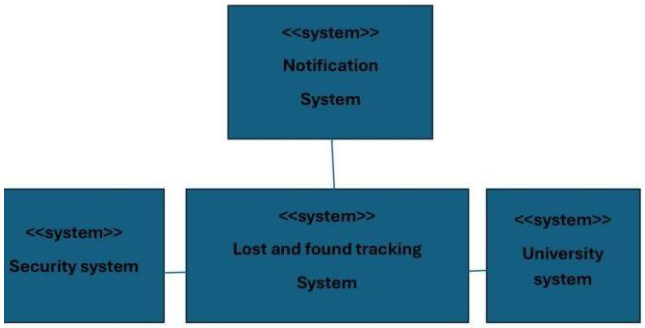
1. User requirement: The system should be reliable system requirement.
   1.1 The system must maintain a high level of availability to make sure that the users can access the lost and found system at any time without interpretation so in order to achieve that, the system must maintain 99.9% uptime monthly so the system can be down for no more than 43 minutes per month.

2. Performance Requirement: The system shall ensure fast response times to enhance user experience.
   2.1 System Requirement: The system must respond to user requests within 2 seconds for 95% of the actions (e.g., searching, reporting, and deleting items). In case of large-scale searches, the system should still complete the operation within 5 seconds.

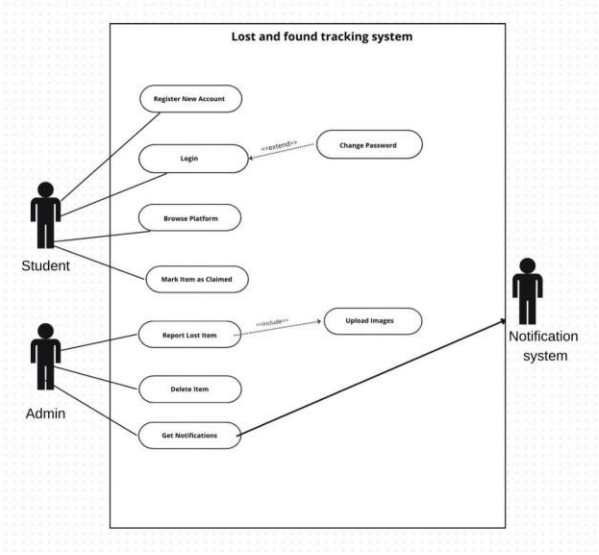
3. Scalability Requirement: The system shall be capable of handling an increasing number of users and data without significant performance degradation.
   3.1 System Requirement: The system must be designed to support up to 1 million active users without performance issues, ensuring that response times remain within 5 seconds for most user interactions.
   3.2 System Requirement: The system must scale horizontally, meaning that additional servers or cloud resources can be added as needed to handle increased traffic.
   3.3 System Requirement: The system must support data growth, ensuring that database performance does not degrade as the amount of stored data increases over time.

2. Chapter 2

Context Diagram



System Use-case Diagram with description



The Lost and Found Tracking System helps users manage lost and found items through two main roles: Student and Admin, along with an external Notification System.

The Student starts with limited access and can perform basic actions. They can Register New Account to create a profile, then Login to access the system. The Login use case is extended by Change Password, meaning changing the password is an optional action available during login. After logging in, the student can Browse Platform to view listed items and Mark Item as Claimed when retrieving a lost item.

Students do not have access to reporting features. Once a student initiates any action related to reporting—such as accessing the reporting interface—they are automatically promoted to Admin, gaining additional privileges.

As an Admin, the user can Report Lost Item, which includes Upload Images as a required step to complete the report. They can also Delete Item to remove outdated entries and Get Notifications to stay updated. The Get Notifications use case is connected to the external Notification System, which delivers alerts and updates.

This structure ensures that optional actions like password changes are handled through «extend», while mandatory steps like uploading images during item reporting are enforced via «include». The automatic role transition from Student to Admin supports efficient item tracking and system control.

Use case scenario (MSS +Extensions)

Use case 1Register New Account

Use-case Name	Register New Account	
Goal	Allows a new user to create an account by providing name, email, and password.	
Actors	Student	
Main Success Scenario	**1**	User opens registration page.
	2	Enters name, email, and password.
	3	System validates input.
	4	System creates account.
	5	System shows confirmation message.
Extension	**2.a**	Invalid email format. The system displays an error message.
	2.b	Email already exists. The system notifies the user.

Use case 2 Login

Use-case Name	Login	
Goal	Allows a user to log in using their credentials.	
Actors	Student	
Main Success Scenario	**1**	1 User opens login page.
	2	Enters email/username and password.
	3	System authenticates credentials.
	4	System opens main interface.
Extension	**2.a**	Invalid credentials. The system shows an error message.

Use case 3 Change Password

Use-case Name	Change Password	
Goal	Allows a user to change their password when needed.	
Actors	Student	
Main Success Scenario	**1**	User requests password change.
	2	System verifies identity.
	3	User enters new password.
	4	System updates password and confirms.
Extension	**2.a**	Verification fails. The user must retry.

	2.b	Weak password. The user is asked to re-enter a stronger password.

Use case 4 Browse Platform

Use-case Name	Browse Platform	
Goal	Allows users to view lost and found items with search and filters.	
Actors	Student	
Main Success Scenario	**1**	User opens the browse page.
	2	System displays lists of lost and found items with filters and search.
	3	User selects an item to view details.
Extension	2.a	No matching results found. The system displays a "No results found" message.

Use case 5 Mark Item as Claimed

Use-case Name	Mark Item as Claimed	
Goal	Allows the reporter to mark an item as claimed.	
Actors	Student	
Main Success Scenario	**1**	User opens the item page
	2	Clicks 'Mark as Claimed'.
	3	System verifies permission
	4	System updates the status and records date/time.
Extension	2.a	Unauthorized user. The system denies the action.
	2.b	User can unmark the item after confirmation.

Use case 6 Report Lost Item

Use-case Name	Report Lost Item	
Goal	Allows the admin to report a lost item with details and an image.	
Actors	Admin	
Main Success Scenario	**1**	Admin opens the 'Report Lost Item' form.
	2	Enters details and uploads an image.
	3	System validates the data.
	4	System saves the report
	5	System confirms success.
Extension	2.a	A required field is missing. The system displays an error message.
	2.b	Image upload fails. The admin can retry.

Use case 7 Upload Images

Use-case Name	Upload Images	
Goal	Allows the admin to upload an image for a lost/found item.	
Actors	Admin	
Main Success Scenario	**1**	Admin selects an image file.
	2	System checks the file type and size.
	3	System uploads and stores the image
Extension	**2.a**	Invalid file. The system displays an error message.
	2.b	Connection issue. The admin can retry.

Use case 8 Delete Item

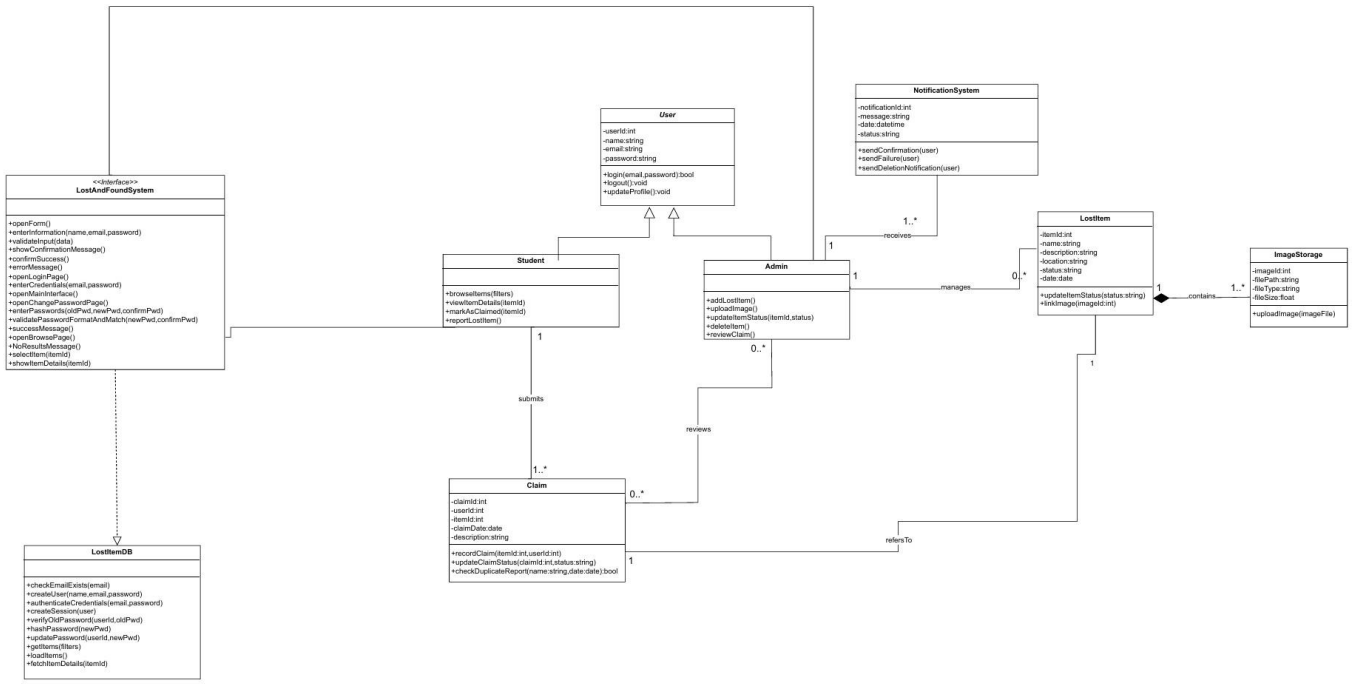
Use-case Name	Delete Item	
Goal	Allows the admin to delete a returned item.	
Actors	Admin	
Main Success Scenario	**1**	Admin opens the item page.
	2	Clicks 'Delete'.
	3	System asks for confirmation.
	4	System deletes and confirms success.
Extension	**2.a**	Deletion is cancelled. No action is taken.

Use case 9 Get Notifications

Use-case Name	Get Notifications	
Goal	Allows the admin to receive notifications via the external Notification System when relevant events occur.	
Actors	Admin (interacts with Notification System)	
Main Success Scenario	**1**	A trigger event occurs.
	2	System generates a notification and stores it.
	3	System delivers it via the Notification System.
	4	Admin views the notification.
Extension	**2.a**	Duplicate event detected. The system ignores it.
	2.b	External delivery fails. The system retries later.

3. Chapter 3

System Class Diagram



System Class Diagram Description

The Lost and Found Tracking System is composed of several interconnected classes that represent the system's core structure and functionality. The system includes two main user roles, Student and Admin, both of which inherit from the User superclass. The User class defines the shared attributes userId, name, email, and password, and contains the methods login, logout, and updateProfile. The Student subclass represents regular users who can interact with the system through the methods browseItems, viewItemDetails, markAsClaimed, and reportLostItem. The Admin subclass extends these capabilities, providing full management privileges to maintain the platform's data through the methods addLostItem, uploadImage, updateItemStatus, deleteItem, and reviewClaim.

The LostItem class represents each recorded item in the system and includes the attributes itemId, name, description, location, status, and date. It provides the methods updateItemStatus and linkImage to modify and associate items with images. Each item can have related images that are managed by the ImageStorage class, which includes the attributes imageId, filePath, fileType, and fileSize, and the method uploadImage for storing item images. The Claim class records details about users claiming or reporting items, containing the attributes claimId, userId, itemId, claimDate, and description, along with the methods recordClaim, updateClaimStatus, and checkDuplicateReport to handle claim operations.

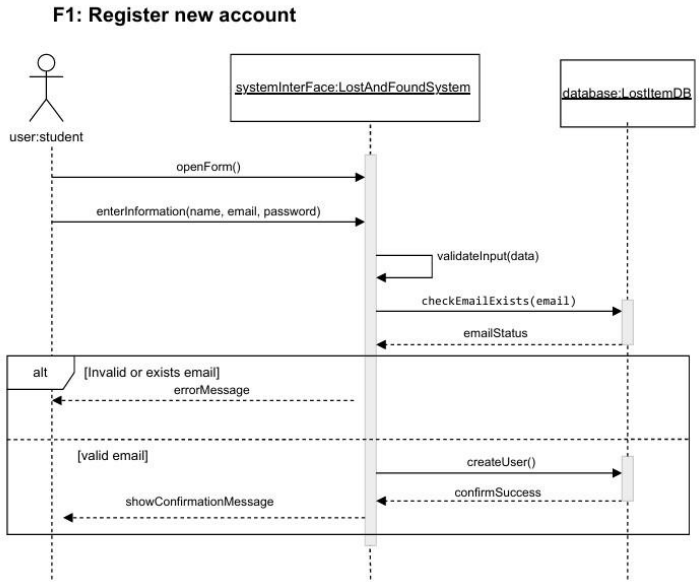
The LostItemDB class is responsible for backend operations and database management. It handles user authentication, password management, and data retrieval through the methods checkEmailExists, createUser, authenticateCredentials, createSession, verifyOldPassword, hashPassword, updatePassword, getItems, loadItems, and fetchItemDetails. The NotificationSystem class functions as an external component that communicates with users by sending alerts and messages. It includes the attributes notificationId, message, date, and status, and provides the methods sendConfirmation, sendFailure, and sendDeletionNotification to deliver notifications and system updates.

The LostAndFoundSystem interface acts as the main boundary between users and the system, managing navigation, form handling, and input validation. It provides several methods such as openForm, enterInformation, validateInput, showConfirmationMessage, confirmSuccess, errorMessage, openLoginPage, enterCredentials, openMainInterface, openChangePasswordPage, enterPasswords, validatePasswordFormatAndMatch, successMessage, openBrowsePage, NoResultsMessage, selectItem, and showItemDetails.

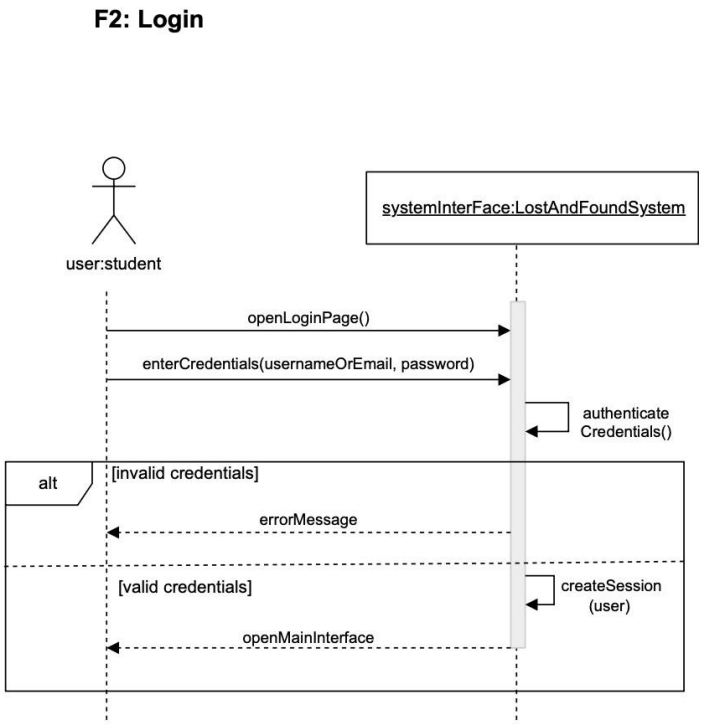
In terms of relationships, Student and Admin inherit from User. Each User can submit multiple Claims, and each Claimis linked to a single LostItem. Each LostItem may have multiple related ImageStorage entries, and each Adminmanages multiple LostItem records. The NotificationSystem sends updates and alerts to users. This design clearly illustrates inheritance, association, and composition relationships, providing a well-structured and maintainable framework for managing lost and found items efficiently. Sequence Diagrams.

Sequence Diagrams

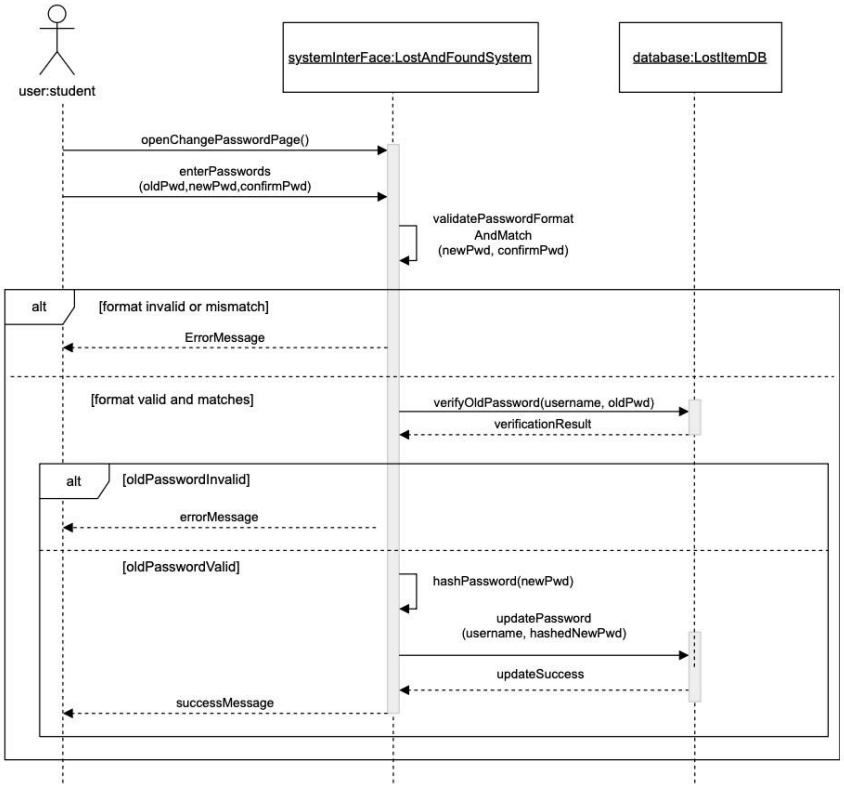
Sequence diagram for REGISTER NEW ACCOUNT



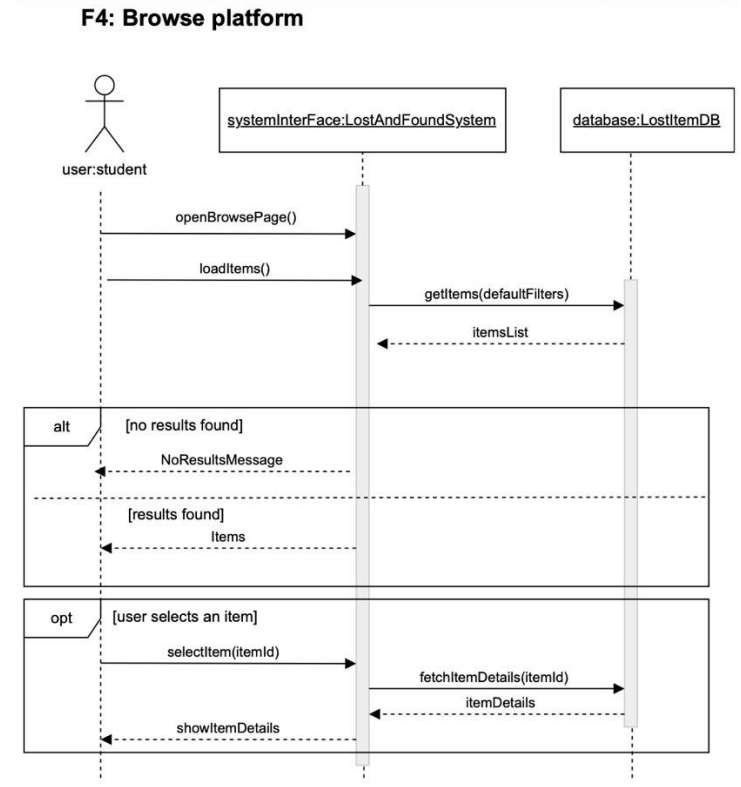
Sequence diagram for LOGIN



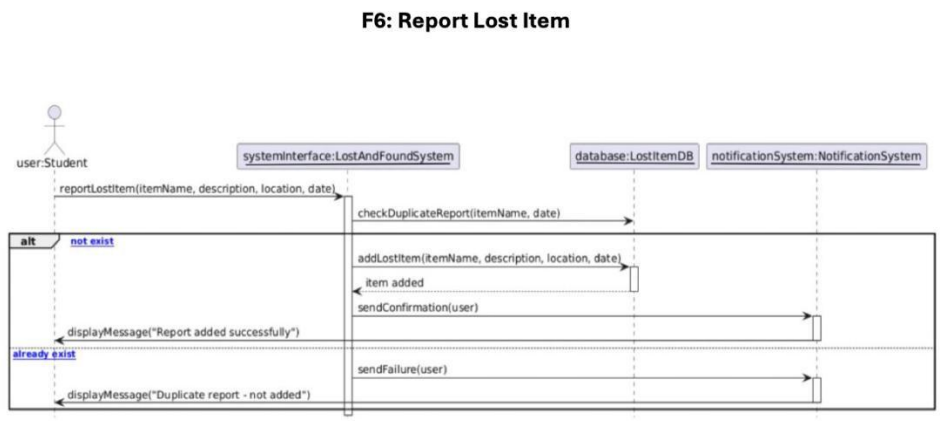
Sequence diagram for CHANGE PASSWORD



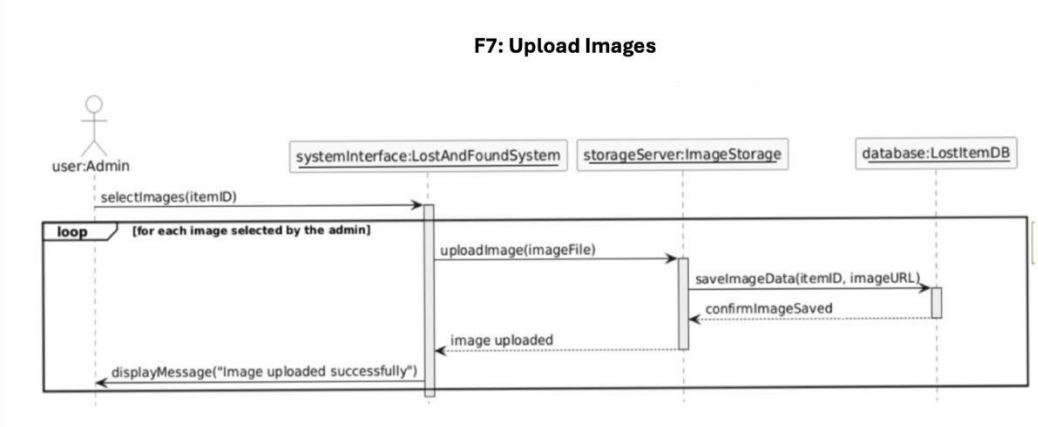
Sequence diagram for BROWSE PLATFORM



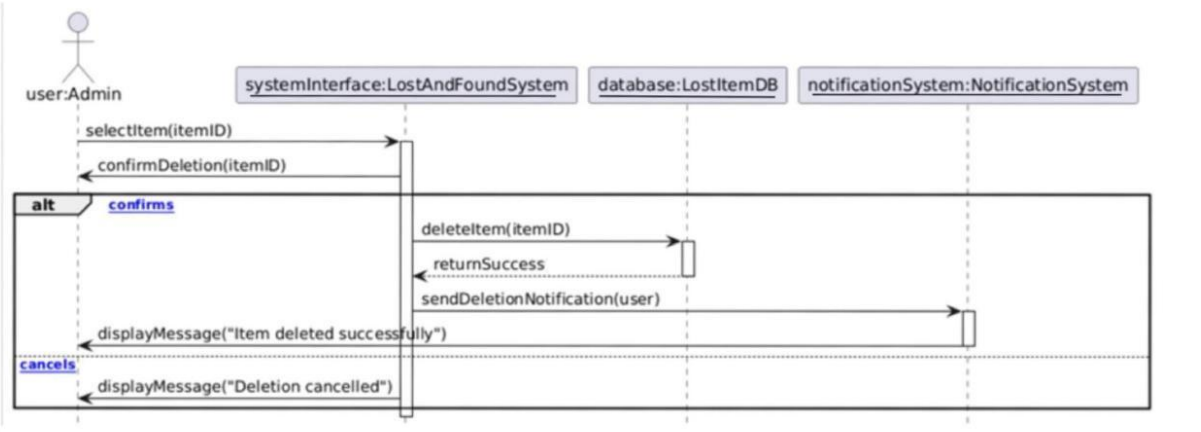
Sequence diagram for REPORT LOST ITEM



Sequence diagram for UPLOAD IMAGES



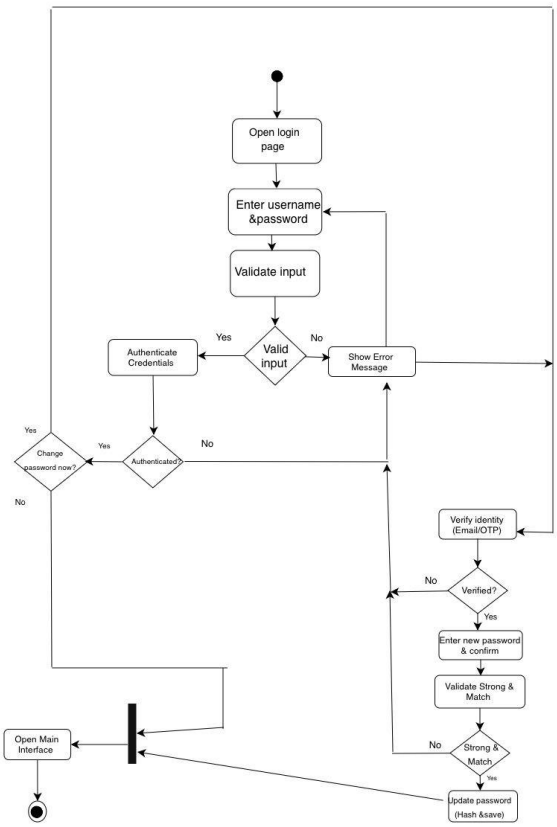
Sequence diagram for DELETE ITEM



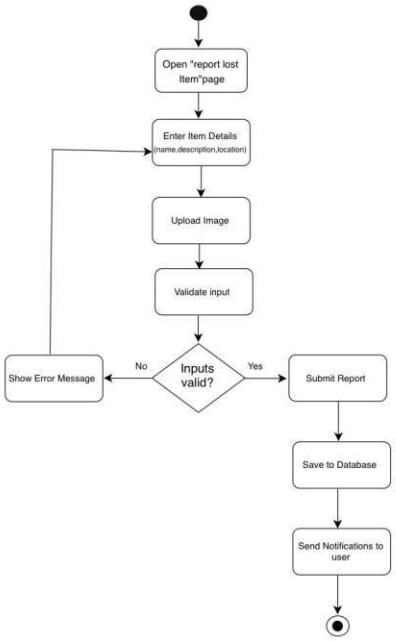
4. Chapter 4

Activity Diagram [one per student]

Activity Diagram for Login + Change Password

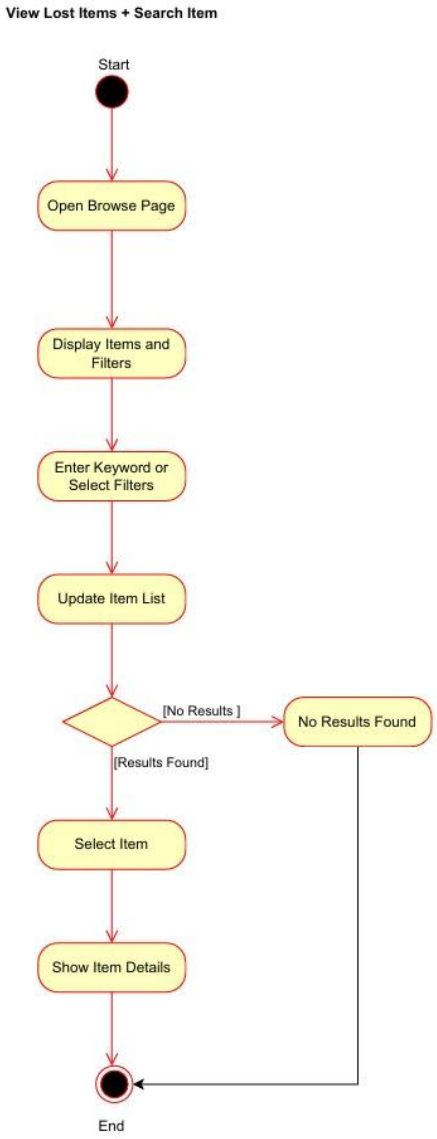


Activity Diagram for Report Lost Item + Upload Image + Get Notification



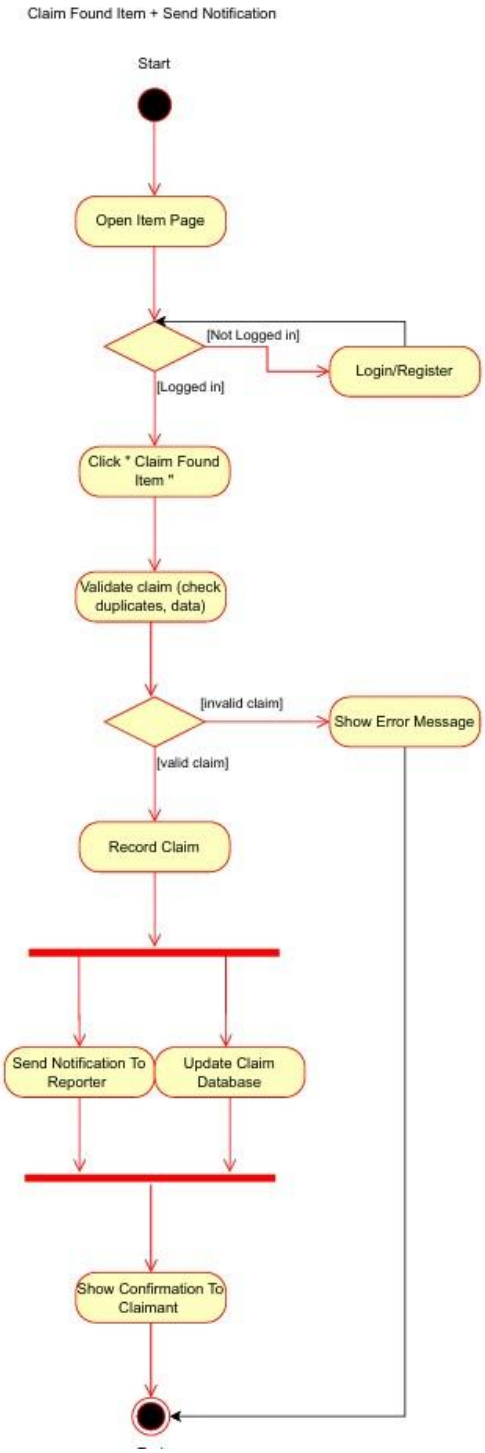
Activity Diagram for

View Lost Items + Search Item



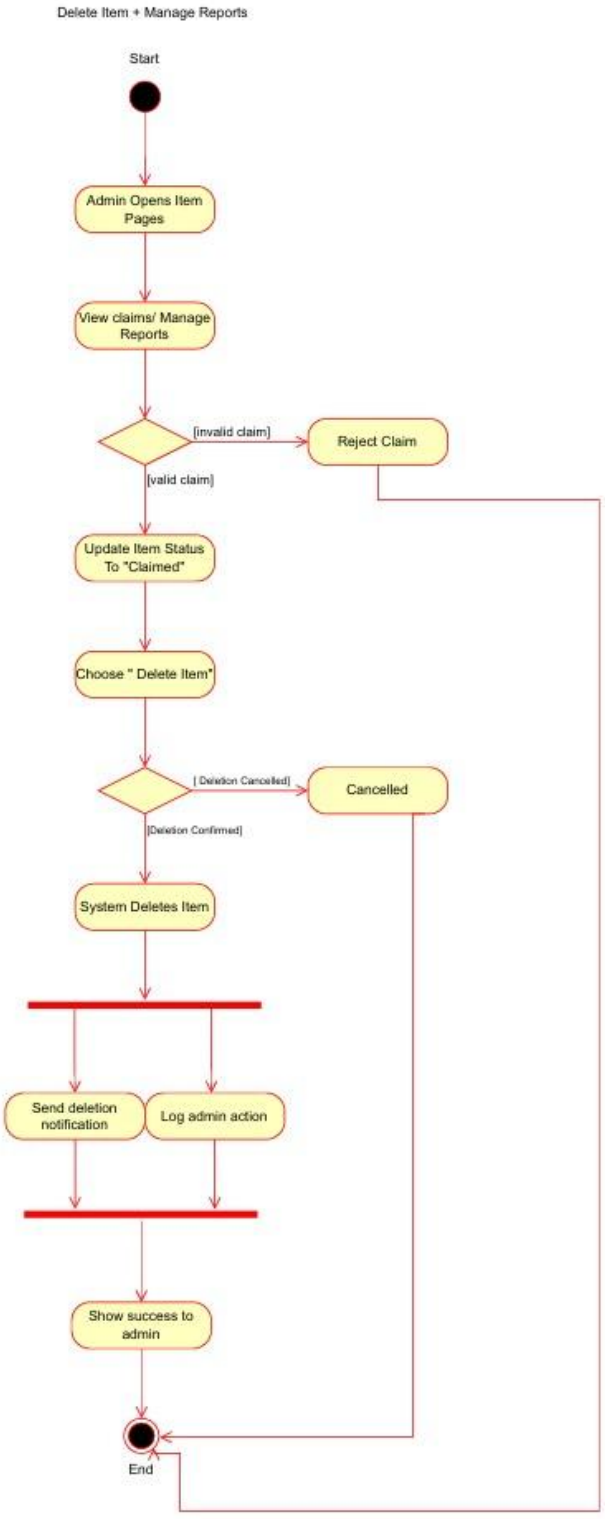
Activity Diagram for

Claim Found Item + Send Notification

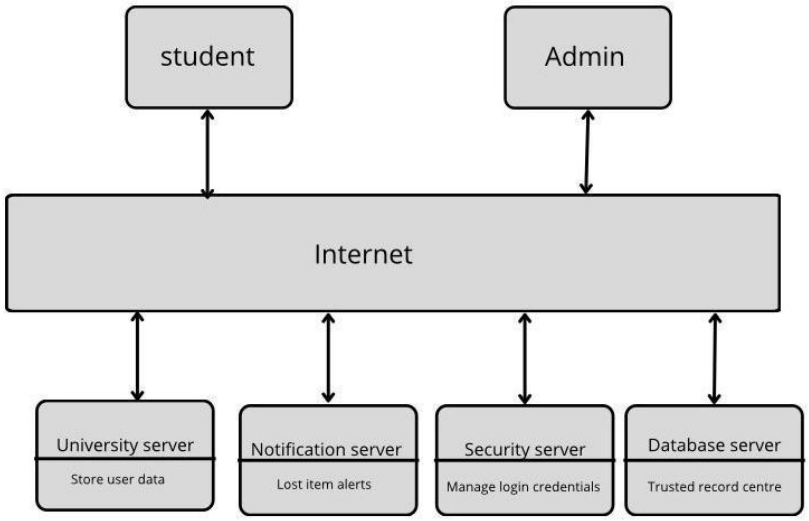


Activity Diagram for

Delete Item + Manage Reports



5. Chapter 5



Architecture Diagram

We choose Client-server model because this architecture allows students and admins to interact with the system efficiently since this model allows students and admins to access the servers from a wide range of locations at any time so even if the student isn't around campus he can still access the servers.

6. Chapter 6

Reference

Sommerville, I. (2011) *Software Engineering*. 9th ed. Pearson Education. Available at: https://engineering.futureuniversity.com/BOOKS%20FOR%20IT/Software-Engineering-9th-Edition-by-IanSommerville.pdf (Accessed: 12 November 2025).